M 7

**Surname** ______________________ **Initials** ______________________

*Van* *Voorletters*

**Stud. No.** ______________________

The undersigned hereby confirms that they are familiar with and accepts the examination regulations of the University of Pretoria.
Your cheat sheet must be handed in with this question paper.

*Die ondergetekende bevestig hiermee dat hulle bewus is van die eksamenregulasies van die Universiteit van Pretoria en dat hy/sy hom/haar daaraan onderwerp. Jy moet jou formuleblad inhandig saam met die vraestel.*

Signature [redacted] *Handtekening* Date 11 / 04 / 2014 *Datum*

**Class Test 3** **11 / 04 / 2014** ***Klastoets 3***

**FSK116**

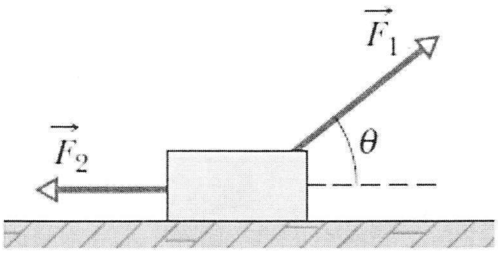
Total/*Totaal*: 22

**NOTE:** Where necessary draw the free body diagrams and use symbols until the second last step where you can then substitute values to obtain the final answer. Remember to use units.
***NEEM KENNIS:*** *Teken die vry-liggaam diagramme waar nodig en gebruik simbole tot die tweede-laaste stap wanneer jy die waardes kan invervang om jou finale antwoord te bereken. Onthou om eenhede te gebruik.*

---

## Question 1/*Vraag 1*

In the Figure, forces $\vec{F}_1$ and $\vec{F}_2$ are applied to a lunchbox as it slides at constant velocity over a frictionless floor. If we are to decrease angle $\theta$ without changing the magnitude of $\vec{F}_1$, should we increase, decrease, or maintain the magnitude of $\vec{F}_2$, in order to keep the lunchbox sliding at a constant velocity? Substantiate your answer.

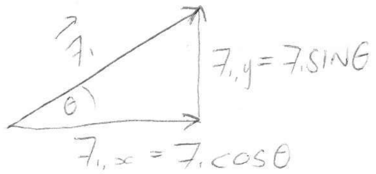



*Die figuur dui twee kragte, $\vec{F}_1$ en $\vec{F}_2$, aan wat toegepas word op 'n kosblik terwyl dit teen 'n konstante snelheid oor 'n wrywinglose vloer gly. Indien ons die hoek $\theta$ kleiner maak sonder om die grootte van $\vec{F}_1$ te verander, moet ons $\vec{F}_2$ groter maak, kleiner maak of dieselfde hou indien ons die kosblik nog steeds teen 'n konstante snelheid wil laat gly? Gee redes vir jou antwoord.*

[3]

$\vec{F}_1$ CAN BE BROKEN UP INTO A HORIZONTAL ($\vec{F}_{1,x}$) AND VERTICAL ($\vec{F}_{1,y}$) COMPONENTS (SEE SKETCH BELOW). WHEN $\theta$ DECREASES, $\vec{F}_{1,x}$ WILL BECOME LARGER AND $\vec{F}_{1,y}$ SMALLER. THIS MEANS THAT $\vec{F}_{1,x}$ WILL HAVE A BIGGER EFFECT ON THE HORIZONTAL MOTION OF THE LUNCHBOX IN THE DIRECTION OF $\vec{F}_{1,x}$. (* NOTE THAT IF $F_{1,y}$ DECREASES THE HORIZONTAL FRICTION ($f_k$) WILL ALSO INCREASE, THIS IS IN THE OPPOSITE DIRECTION OF MOTION)

2




FOR THE LUNCHBOX TO KEEP SLIDING AT CONSTANT VELOCITY, $\vec{F}_2$ (WHICH HAS ONLY A HORIZONTAL COMPONENT) MUST BE INCREASED TO BALANCE THE EFFECTS OF THE CHANGES IN $\vec{F}_{1,x}$ AND FRICTION $f_K$.

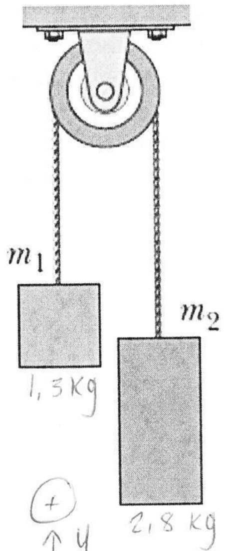
2

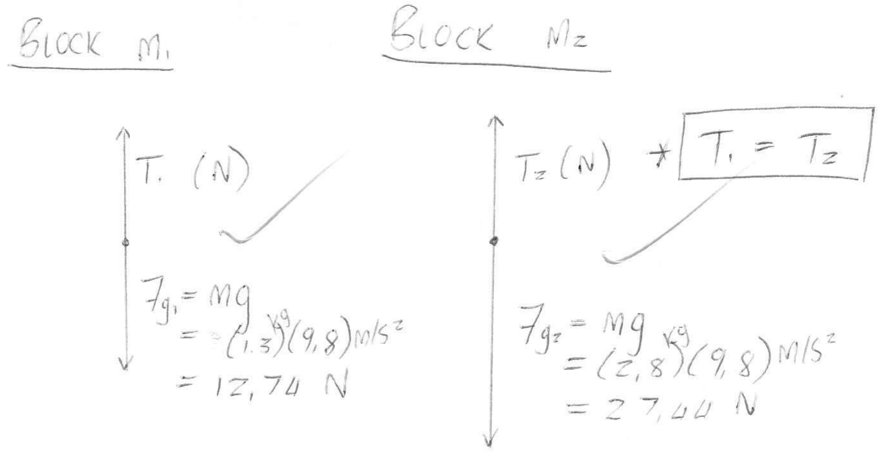
**Question 2/*Vraag 2***

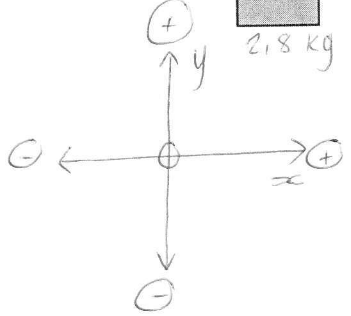
An *Atwood Machine*, consisting of two blocks, are connected by a cord of negligible mass that passes over a frictionless pulley (also of negligible mass). One block has a mass $m_1$ = 1.30 kg; the other has a mass $m_2$ = 2.80 kg. Determine the acceleration of the blocks and the tension in the cord.

*'n Atwood-masjien bestaan uit twee blokke wat gekoppel is aan mekaar deur middel van 'n tou (van weglaatbare massa) wat oor 'n wrywinglose katrol (ook van weglaatbare massa) hang. Een blok het 'n massa $m_1$ = 1.30 kg; die ander 'n massa $m_2$ = 2.80 kg. Bepaal die versnelling van die blokke en die spankrag in die tou.*

[2+6]




$F_{NET} = ma$ (NEWTON'S SECOND LAW)

FOR BLOCK $M_1$: $F_{NET} = ma$

$T - F_{g_1} = ma$

$T - (12,74\text{ N}) = (1.3\text{ kg})a$ ... (A)

FOR BLOCK $M_2$: $F_{NET} = ma$

$T - F_{g_2} = ma$ ✗

$T - (27,44\text{ N}) = (2,8\text{ kg})a$

$T = (2,8\text{ kg})a + (27,44\text{ N})$ ... (B)

SUBSTITUTE (B) INTO (A)

$[(2,8\text{ kg})a + (27,44\text{ N})] - (12,74\text{ N}) = (1,3\text{ kg})a$

$(2,8\text{ kg})a - (1,3\text{ kg})a = (12,74\text{ N}) - (27,44\text{ N})$

$(1,5\text{ kg})a = (-14,7\text{ N})$

$a = -9,8\text{ m/s}^2$ ... (C) $\therefore$ 9,8 m/s$^2$ ✗ DOWNWARDS

SUBSTITUTE (C) INTO (A)

$T - (12,74\text{ N}) = (1.3\text{ kg})(-9,8\text{ m/s}^2)$

$T = 0$ N ✗

3



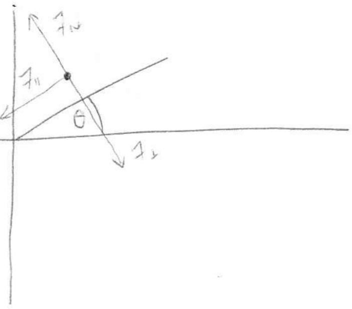
3

**Question 3/*Vraag 3***

a) A banked circular highway curve is designed for traffic moving at 65 km/h to move through the curve without relying on the friction between the tyres and the dry road surface. The radius of the curve is 200 m. Determine the angle at which the highway surface must be banked (tilted) with respect to the horizontal.
*'n Sirkelvormige draai op 'n snelweg is op 'n skuinste ontwerp sodat verkeer wat teen 65 km/h deur die draai beweeg nie op die wrywing tussen die bande en 'n droë padoppervlak hoef staat te maak nie. Die radius van die draai is 200 m. Bepaal die hoek waarteen die snelweg oppervlak gekantel moet word ten opsigte van die horisontale rigting.*

[2+4]

$R = 200\ m$




$$V = 65\ km/h = \frac{(65)(1000)}{(60)(60)}\ \frac{m}{s} = \underline{18{,}05\ m/s}$$

$F_{NET} = ma$ (NEWTON'S SECOND LAW)

FOR $x$: $F_{NET} = ma$

$+F_{\parallel} = (m)\left(\frac{V^2}{R}\right) \ldots$ (A) $\quad$ * $a = \frac{V^2}{R}$

FOR $y$: $F_{NET} = ma$

$F_N - F_{\perp} = m(0)$ $\quad$ * NO ACCELERATION VERTICALLY

(A) $\rightarrow$ $+F_{\parallel} = (m)\left(\frac{V^2}{R}\right)$ $\quad$ * $F_{\parallel} = mg\sin\theta$ (SEE NOTE ↙)

$+\cancel{m}g\sin\theta = (\cancel{m})\left(\frac{V^2}{R}\right)$ ✓ $\quad$ * m BOTH SIDES OF =

$+g\sin\theta = \frac{V^2}{R} \ldots$ (B)

NOTE:

* $F_{\parallel}$ - COMPONENT OF $F_g$ ~~ALO~~ PARALLEL TO SURFACE
$F_{\parallel} = mg\sin\theta$
* $F_{\perp}$ - COMPONENT OF $F_g$ PERPENDICULAR TO SURFACE
$F_{\perp} = mg\cos\theta$

(B) $\rightarrow$ $\sin\theta = +\frac{V^2}{(R)(g)}$

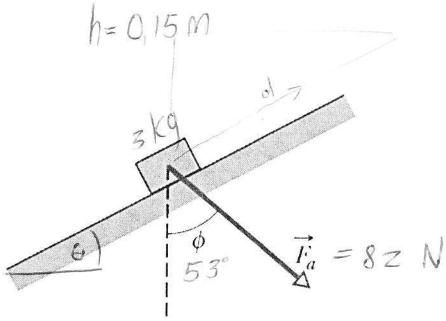
$$\sin\theta = +\frac{(18{,}05)^2}{(200)(9{,}8)}$$

$$\theta = +9{,}57° \quad ✗$$

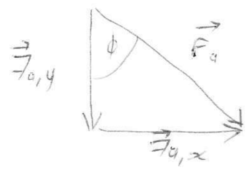
**Question 4/*Vraag 4***

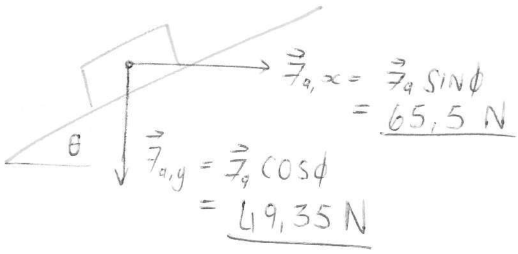
In the figure, a constant force $\vec{F}_a$ of magnitude 82.0 N is applied to a 3.00 kg shoe box at an angle $\phi = 53.0°$, as shown, causing the box to move **UP** a frictionless ramp at a **constant speed**. Calculate how much work is done on the box by $\vec{F}_a$ when the box has moved through a vertical distance $h = 0.150$ m upwards.

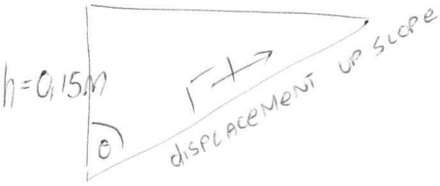
*Die figuur dui 'n krag $\vec{F}_a$, met grootte 82.0 N aan, wat inwerk op 'n 3.0 kg skoenboks teen 'n hoek $\phi = 53.0°$, soos aangdui. Dit veroorsaak dat die skoenboks by die wrywinglose skuinste* ***OP*** *beweeg teen 'n* ***konstante spoed****. Bereken die arbeid verrig op die skoenboks deur die krag $\vec{F}_a$ as die boks deur 'n vertikale afstand h = 0.150 m opwaarts beweeg het.*




[5]




$W = Fd\cos\phi$

$a = 0$ (CONSTANT SPEED)

$F_{NET} = ma$ (NEWTONS SECOND LAW)

$F_{NET} = (3)(0)$

$W = Fd\cos\phi$ ✓

$W = (82)$(DISPLACEMENT UP SLOPE (M)) cos (ANGLE BETWEEN $F_a$ AND DISPLACEMENT)

$W = Fd\cos\phi$

$= (82\text{ N})(0{,}15\text{ m})\cos(180° - 53°) =$

$= (82\text{ N})(0{,}15\text{ m})\cos(127°)$

$= -7{,}4$ J ✗

# THE END / DIE EINDE